WALMARTSALESPERFORMANCEDASHBOARD

Using Microsoft Power BI Project Report

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Tool Used: Microsoft Power BI Desktop

1. Introduction

For this project, I built a **Sales Performance Dashboard in Power BI** using Walmart's weekly sales data. I selected this dataset because it contains a good mix of numerical and categorical information, including store-level sales, dates, fuel prices, CPI, unemployment, and holiday indicators. This gave me the opportunity to explore different types of visualizations and understand how various factors relate to sales performance.

The main purpose of the dashboard was to transform a large and unfiltered dataset into a **simple and interactive visual report** that a manager could understand quickly. The dashboard allows users to filter the data by year, store, and holiday status and instantly observe how these filters affect the sales figures.

Visual charts are important for tracking sales performance because they make large amounts of data easier to understand and help identify trends, comparisons, and changes quickly. In this project, the charts are interconnected, so when a filter or selection is applied, the related visuals update automatically. This makes the dashboard interactive rather than just a collection of static charts.

This is my first visual chart, and it provides an initial view of the sales data and helps in understanding the overall sales performance before moving on to deeper analysis.

2. Objectives

Coming up with the project, among others, I had the following objectives in mind that the dashboard should meet:

1. To conduct analysis of Walmart’s weekly sales at all 45 stores to find out the general tendency in sales.

2. To identify KPIs, including total sales, average weekly sales, number of transactions.

3. To make an interactive dashboard, from which it should be possible to navigate to the needed variable and make filters and selections.

4. To compare the sales for holiday and non-holiday weeks and find out if there is a tendency for changes in sales in these two types of weeks.

5. To track tendencies in sales over time.

6. To assess the connection, if any, between the fuel prices and sales for the same week.

7. To present the information in interconnected graphs to provide a better perception of the data and identify key trends.

3. Dataset Description

The dataset I used contains weekly sales records collected across Walmart's stores. The fields I actually worked with were:

● Store—storeidentifier (1 to 45)

● Date—theweekthesaleswererecorded

● WeeklySales—total sales for that store in that week

● FuelPrice—averagefuel price for that week

● CPI—ConsumerPriceIndex

● Unemployment—unemploymentrate

● HolidayFlag—whetherthat weekincluded aholiday

4. Data Preparation

Before I got into building any visuals, I spent some time cleaning up the dataset. The Date column, for instance, wasn't recognized as a proper date type when I first imported it, so I had to fix that in Power Query before anything downstream would work correctly. Here's roughly what I did:

● ImportedtheExcelfile into Power BI Desktop.

● Correctedthe date format so Power BI would readit as a proper date field.

● CheckedthatWeeklySales, Fuel Price, CPI, and Unemployment were all recognized as numeric fields.

● SetuptheDAXmeasuresneededtodrivethe KPI cards.

5. DAX Measures

Used I created four core measures to power the KPI cards at the top of the dashboard:

● TotalSales

● AverageSales

● StoreCount

● MaximumSales

These were simple SUM, AVERAGE, DISTINCTCOUNT, and MAX measures built off the Weekly Sales column, but they're what make the top row of the dashboard readable at a glance.

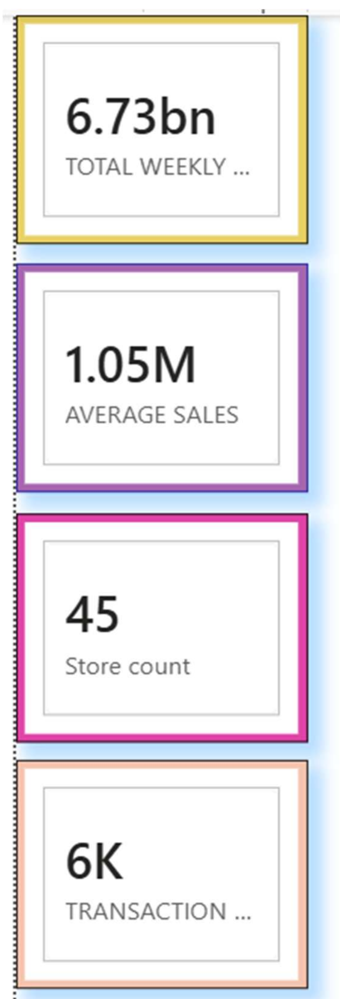
6. Data Components and Visuals Used

After importing the Walmart weekly sales dataset into Power BI, I created different visual components to analyse the sales performance. I selected each visual based on what I wanted to understand from the data. I also created measures for the important calculations and connected the visuals so that they respond to the filters applied on the dashboard.

6.1 KPI Cards

First, I created **four KPI cards** to give a quick summary of the dataset. For these cards, I created and used the following measures:

- **Total Sales:** I used the **TOTAL WEEKLY SALES** measure to calculate the overall sales value.
- **Average Sales:** I used the **AVERAGE SALES** measure to calculate the average weekly sales.
- **Transaction Count:** I used the **TRANSACTION COUNT** measure to show the total number of transactions.
- **Store Count:** I used the **Store count** measure to show the number of stores included in the analysis.



I placed these measures in separate cards so that the main performance figures could be understood immediately without going through the detailed charts.

6.2 Summary Table

Next, I created a **summary table** to look at the performance of individual stores. I used **Store** as the main field and added **Total Sales, Average Sales, and Transaction Count** as the measures.

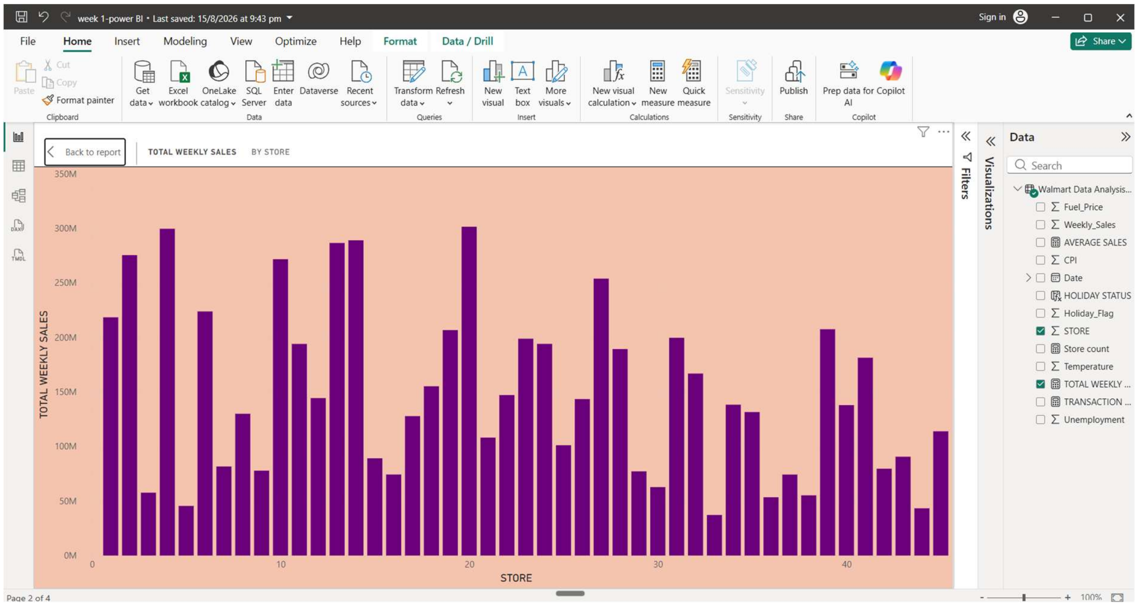
I created this table because the KPI cards only provide an overall summary, whereas the table allows me to compare the performance of individual stores in more detail.

Calculations Calendars

STORE	TOTAL WEEKLY SALES	AVERAGE SALES	TRANSACTION COU
1	21,83,65,470.21	15,37,785.00	
2	27,53,82,440.98	19,25,751.34	
3	5,75,86,735.07	4,02,704.44	
4	29,95,43,953.38	20,94,712.96	
5	4,54,75,688.90	3,18,011.81	
6	22,37,56,130.64	15,64,728.19	
7	8,15,98,275.14	5,70,617.31	
8	12,99,51,181.13	9,08,749.52	
9	7,77,89,218.99	5,43,980.55	
10	27,16,17,713.89	18,99,424.57	
11	19,39,62,786.80	13,56,383.12	
12	14,42,87,230.15	10,09,001.61	
13	28,65,17,703.80	20,03,620.31	
14	28,89,99,911.34	20,20,978.40	
15	8,91,33,683.92	6,23,312.47	
16	7,42,52,425.40	5,19,247.73	
17	12,77,82,138.83	8,93,581.39	
18	15,51,14,734.21	10,84,718.42	
19	20,66,34,862.10	14,44,999.04	
20	30,13,97,792.46	21,07,676.87	
21	10,81,17,878.92	7,56,069.08	
22	14,70,75,648.57	10,28,501.04	
23	19,87,50,617.85	13,89,864.46	
24	19,40,16,021.28	13,56,755.39	
25	10,10,61,179.17	7,06,721.53	
26	14,34,16,393.79	10,02,911.84	
Total	**6,73,46,60,055.43**	**10,46,404.61**	**6**

6.3 Total Weekly Sales by Store

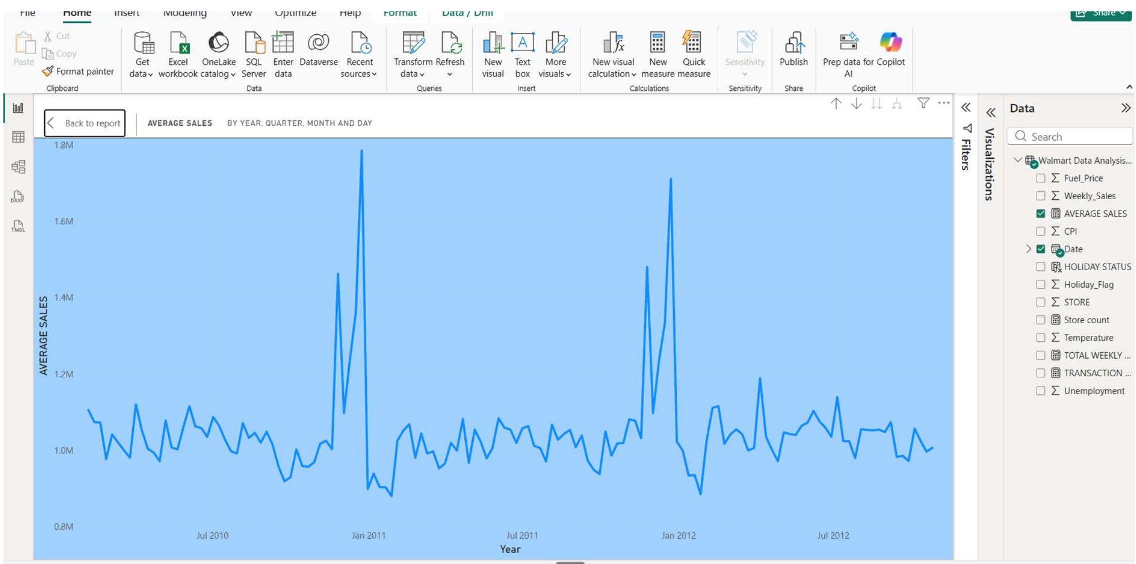
For comparing sales between stores, I created a **Clustered Column Chart**. I placed **Store** on the axis and used **TOTAL WEEKLY SALES** as the value.



This helped me compare the total sales generated by each store and identify which stores had relatively higher or lower sales.

6.4 Average Sales Trend Over Time

To understand how sales changed over time, I created a **Line Chart**. I used the **Date hierarchy** on the time axis and **AVERAGE SALES** as the value.



I used the Date hierarchy so that I could analyse the sales trend at different levels such as year, quarter, month, and day. This helped me observe changes in average sales over the selected period.

6.5 Average Sales by Store

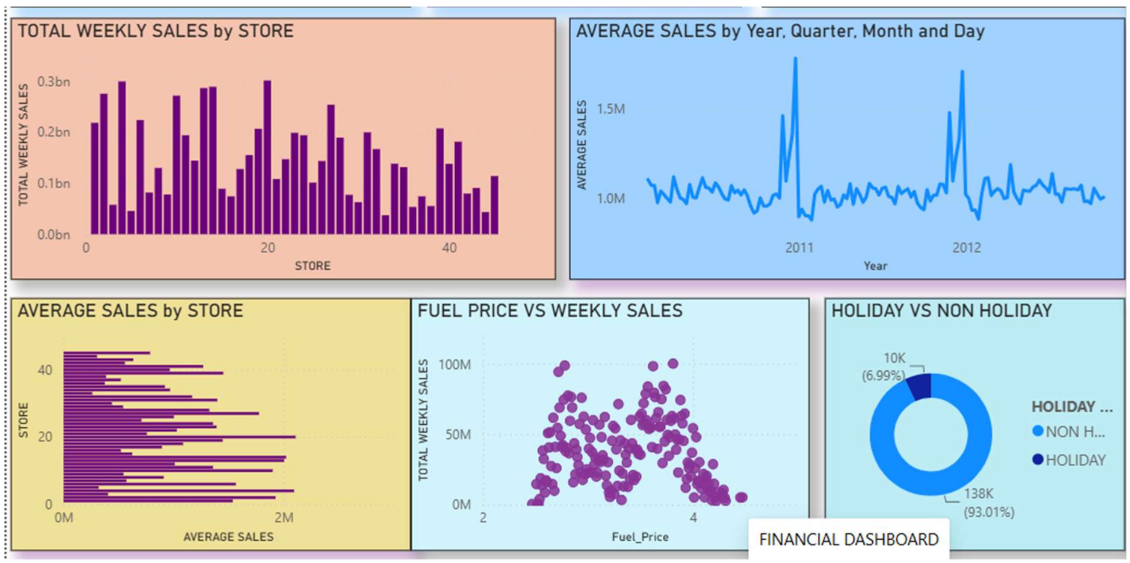
I also created a **Clustered Bar Chart** to compare the average sales of different stores. I used **Store** as the category and **AVERAGE SALES** as the value.

This visual helped me understand which stores had higher or lower average sales rather than looking only at their total sales.

6.6 Cumulative Weekly Trend

To show the movement of sales over time, I created an **Area Chart** titled **"CUMULATIVE WEEKLY TREND."** I used the **Date hierarchy** and **TOTAL WEEKLY SALES** for this visual.

I selected an area chart because it gives a clear visual representation of how sales values move across the selected time period and makes the overall trend easier to observe.



6.7 Fuel Price vs Weekly Sales

To check whether fuel prices had any relationship with Walmart's sales, I created a **Scatter Chart** titled **"FUEL PRICE VS WEEKLY SALES."**

I used **Fuel Price** as one variable and **TOTAL WEEKLY SALES** as the other variable. I selected a scatter chart because it is useful for observing the relationship between two numerical variables.

6.8 Holiday vs Non-Holiday

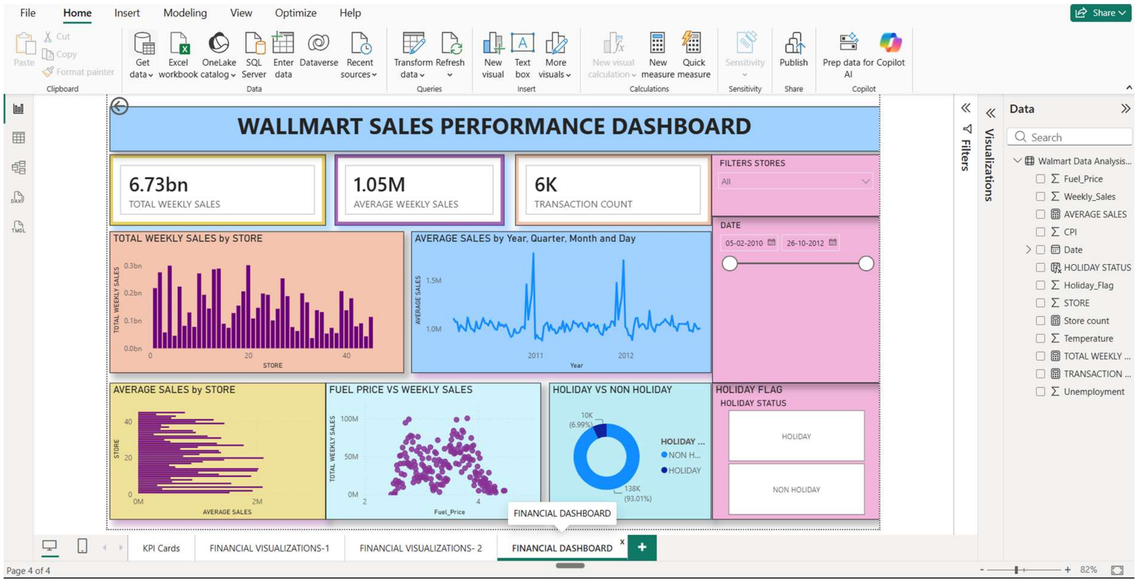
To compare holiday and non-holiday periods, I created a **Donut Chart** titled **"HOLIDAY VS NON HOLIDAY."**

I used **HOLIDAY STATUS** as the category and **Store** as the value. This allowed me to visually compare the distribution between holiday and non-holiday records in the dataset.

6.9 Store Ranking Over Time

I created a **Ribbon Chart** titled **"STORE RANKING OVERTIME"** to understand how the position of different stores changed over time.

For this visual, I used **Store** as the category, **TOTAL WEEKLY SALES** as the value, and the **Date hierarchy** to analyse the changes over time. The ribbon chart helped me observe changes in the relative ranking of stores based on their sales performance.



6.10 Slicers and Interactivity

Finally, I added **three slicers** to make the dashboard interactive:

- **Store:** I used this slicer to select and analyse a particular store.
- **Date:** I used this slicer to select a specific date or time period.
- **Holiday Flag:** I used this slicer to filter the analysis based on holiday status.

The main advantage of using these slicers is that the charts and KPI cards are interconnected. When I select a store, date, or holiday status, the related visuals automatically update. This made the dashboard more interactive and allowed me to analyse the same dataset from different perspectives.

Overall, these visual components helped me move from a **simple summary of the data to a more detailed analysis of sales, stores, time trends, fuel prices, and holiday periods**.

7. Key Findings

1. I found that the dataset contains **6,435 weekly sales records across 45 stores**, giving a broad view of Walmart's sales performance.
2. I observed that the **average weekly sales were approximately $1.047 million** across the dataset.
3. I found that **Store 20 recorded the highest total sales of approximately $301.40 million**.
4. I observed that **Store 33 had the lowest total sales at approximately $37.16 million**, showing a large difference in performance between stores.
5. I found that the highest individual weekly sale was approximately **$3.82 million**, showing a significant peak in sales during the period.
6. I observed that **450 out of 6,435 records were holiday weeks**, which represents approximately **7% of the dataset**.
7. I found that the **average non-holiday weekly sales were approximately $1.041 million**, which helped me compare normal weeks with holiday periods.
8. I observed that **fuel prices ranged from $2.47 to $4.47**, while weekly sales showed only a weak relationship with fuel prices.

CONCLUSION

Through this project, I learned how Power BI can be used to convert raw sales data into an interactive and easy-to-understand dashboard. I created KPI cards, tables, charts, and slicers to analyse Walmart's weekly sales from different perspectives.

The dashboard helped me understand store-wise performance, average sales, sales trends over time, holiday patterns, fuel price relationships, and changes in store rankings. I also understood that visualisation makes it easier to identify patterns and compare large amounts of data compared with looking at raw data alone.

Overall, this project gave me practical experience in selecting suitable visuals, creating measures, using filters, and connecting different charts in Power BI. It also helped me understand how a dashboard can support better data analysis and make the information easier for a user or manager to interpret.